

Probabilistic Classification

2. Conditional / Discriminative (e.g., Logistic Regression)

- Only model conditional probability $P(y/x)$
- Prediction can be done directly:

$$y^* = \operatorname{argmax}_y P(y/x)$$

- Advantages: no need to model $P(x)$, easier to develop feature-rich models for $P(y/x)$

Discriminative Approach

- Where there are complex features, generative approaches are more difficult to use
 - For example, highly correlated features
 - The identity of individual words (unigram features)
 - The identity of sequences of n words in the document (n-gram features)
- Many feature-based discriminative classification techniques out there, but log-linear models are still popular in the NLP community!

Text Classification

- Goal: classify documents into categories

... win the election ... *POLITICS*

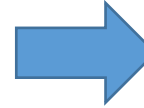
... win the game ... *SPORTS*

... see a movie ... *OTHER*

- Classically: based on bag of words in the document
- But other information sources are potentially relevant: document length, average word length, document's source, document layout

Feature Representation

Washington County jail *served*
11,166 meals last month - a figure
that translates to feeding some 120
people three times daily for 31 days

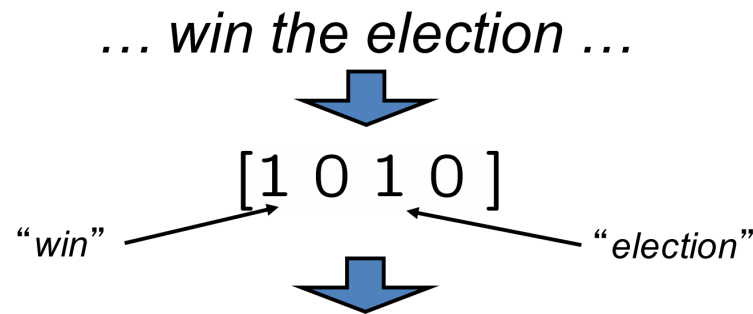


context:jail = 1
context:county = 1
context:feeding = 1
context:game = 0
...
local-context:jail = 1
local-context:meals = 1
...
object-head:meals = 1
object-head:ball = 0

- Features are indicator functions which count the occurrences of certain patterns in the input
- Different feature values for every pair of input x and class y

Block Notation

- We often think of the feature function as a mapping from a pair of input and label pair to a feature vector
 - In these cases, the feature vector will take a block form, as below



$$\begin{aligned}\phi(x, \textit{SPORTS}) &= [1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0] \\ \phi(x, \textit{POLITICS}) &= [0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0] \\ \phi(x, \textit{OTHER}) &= [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0]\end{aligned}$$

Prediction

- In a linear model, each feature gets a weight
 - Feature function: ϕ
 - Weight vector: w

$$\begin{aligned}\phi(x, \textit{SPORTS}) &= [1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0] \\ \phi(x, \textit{POLITICS}) &= [0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0] \\ \hline w &= [1 \ 1 \ -1 \ -2 \ 1 \ -1 \ 1 \ -2 \ -2 \ -1 \ -1 \ 1]\end{aligned}$$

- The prediction of y is the value that maximizes the score

$$\begin{aligned}\phi(x, \textit{SPORTS})^T w &= 1 * 1 + (-1) * 1 = 0 \\ \phi(x, \textit{POLITICS})^T w &= 1 * 1 + 1 * 1 = 2\end{aligned}$$



Prediction(x) = *POLITICS*